TCON UNION '21

ANTIDOTE

ASSORTED PREVIOUS YEAR QUESTION PAPERS

MODULE 1

TCON UNION 2021

COLLEGE OF NURSING
THALASSERY

TCON ACADEMIC CLUB

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ANATOMY

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UNIT-1

INTRODUCTION TO ANATOMICAL TERMS

Short note:

1. Compound epithelium (5)
2. Parts of a cell (5)
3. Human cell (5)
4. Connective tissue cells (5)
5. Hyaline cartilage (5)
6. Types of cartilage (5)

Answer briefly:

1. Simple epithelium (3)
2. Transitional epithelium (3)
3. Haversian system (3)
4. Stratified squamous epithelium(3)
5. Compound epithelium.(3)

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UNIT-2

SKELETAL SYSTEM

Essay:

1. Classify joints and describe shoulder joint under the following headings (10)
   - Articular surfaces - Ligaments - Applied Anatomy
2. Classify bones and describe the different types. Draw and label both the macroscopic and microscopic structure of long bone.(10)
3. Describe the classification of joints. Describe each type with examples. Explain the applied anatomy of synovial joint with suitable examples. (12)

Short essay:

1. Mention the classification of bones and describe the different types. Draw and label structure of a synovial joint.

Short note:

1. Ligaments and muscles supporting shoulder joint (5)
2. Elbow joint (5)
3. Muscles around shoulder joint(5)
4. Structure of compact bone(5)
5. Structure of bone (5)
6. Shoulder joint.(5)

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7. Clavicle(5)
8. Types of bone(5)
9. Classification of joints (5)
10. Synovial joint.(5)
11. Hip joint – movements and muscles producing it*(5)
12. Types ,Classification and Structure of Bones (5)
13. Sternum(5)

Answer briefly:

1. Parts of femur (3)
2. Classification of joints* (3)
3. Upper end of humerus* (3)
4. Parts of humerus(3)
5. Upper end of femur.(3)
6. Parts of bony pelvis(3)
7. Histology of skeletal muscle.(3)

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UNIT-3

MUSCULAR SYSTEM

Essay:

1. Describe the classification of muscles. Describe the structure of one type in detail. Mention the applied anatomy of this type. (12)

Short note:

1. Gluteus Maximus muscle**** (5)
2. Deltoid muscle***** (5)
3. Thigh muscles. (5)
4. Types and structure of muscles(5)
5. Histology of Muscles(5)
6. Adductor muscles of thigh (5)

Answer briefly:

1. Applied aspect of deltoid muscle (3)
2. Deltoid muscle (3)
3. Structure of muscle (3)

Differentiate between:

1. Inversion and eversion movements (4)

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UNIT-4

NERVOUS SYSTEM

Essay:

1. Name the cranial nerves. Describe the facial nerve under the following headings/Course and distribution /Branches /Applied aspects (10)

Short essay:

1. Explain the structure of neuron. Add a note on neuroglia (7)
2. Describe the structure of neuron with the help of a neat labelled diagram. Add a note onneuroglia.* (7)

Short note:

1. Femoral nerve (5)
2. Name the cranial nerves (5)
3. Neuron* (5)
4. Blood supply of cerebrum (5)
5. Parts of brain* (5)
6. List the cranial nerves (5)
7. Spinal cord* (5)
8. Multipolar neuron labelled diagram (5)
9. Blood supply of spinal cord (5)
10. Meninges (5)
11. Supero-lateral surface of cerebral hemisphere (5)
12. Specifications, Parts and Structure of Spinal Cord (5)
13. Ventricles of brain (5)
14. Cerebrum(5)

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Answer briefly:

1. Structure of spinal cord (3)
2. Sciatic nerve*** (3)
3. Enumerate cranial nerves*.(3)
4. Spinal nerve (3)
5. Neuron*(3)
6. Cross section of spinal cord (3)
7. Meninges (3)
8. Axillary nerve*(3)
9. Circle of willis(3)
10. Coverings of brain (3)
11. Fontanelle of fetal skull. (3)
12. Radial nerve (3)

Differentiate between:

1. Central nervous system and peripheral nervous system (4)
2. Cerebrum and cerebellum (4)
3. Sympathetic and parasympathetic nervous system* (4)
4. Cranial nerve and spinal nerve (4)

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UNIT-5

SENSORY ORGANS

Short notes:

1. Layers of eye ball* (5)
2. Structure of eyeball* (5)
3. Structure of middle ear cavity (5)
4. Structure of skin(5)
5. Skin (5)
6. Middle ear.(5)
7. Tympanic membrane (5)

Answer briefly:

1. Contents of middle ear (4)
2. Layers of eyeball* (4)
3. Ear ossicles (4)**
4. Muscles of eye ball (4)*
5. Structure of hairy skin(4)
6. Structure of thick skin(4)
7. Histology of skin(4)

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UNIT-6

CIRCULATORY & LYMPHATIC SYSTEM

Essay:

1. Describe the heart under the following headings:
   Pericardium / Chambers / Blood supply / applied anatomy (10)

Short essay:

1. Draw and label the lymphatic organs. Describe the microscopic structure of one of them. (7)
2. Draw and label the main arteries of the upper limb. Describe the structure of blood vessels (7)
3. Draw and label the chambers and vessels of heart. Describe the interior of heart in detail. (7)

Short note:

1. Right atrium (5)
2. Tonsil (5)
3. Femoral artery (5)
4. Structure of lymph node*(5)
5. Draw and label chambers of heart* (5)
6. Spleen * (5)
7. Coronary arteries.(5)
8. Histology of tonsil(5)
9. Lymph node (5)
10. Palatine tonsil(5)
11. Blood supply of heart(5)

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Answer briefly:

1. Blood supply of heart (4)
2. Radial artery (4)
3. Cephalic vein (4)
4. Ventricles of heart (4)
5. Right atrium*(4)
6. List the major arteries of the lower limb. (4)
7. Chambers of heart (4)
8. Great saphenous vein.**(4)
9. Median cubital vein**(4)
10. Coronary sinus(4)
11. Coronary arteries (4)
12. Pericardium(4)
13. Femoral artery(4)
14. What are the branches of aorta (4)

Differentiate between:

1. Structure of artery and vein (4)
2. Coronary and systemic circulation (4)
3. Coronary and pulmonary circulation (4)
4. Systemic circulation and pulmonary circulation (4)

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UNIT-7

RESPIRATORY SYSTEM

Essays:

1. Describe, draw and label the parts of the respiratory system. Add a note on its applied anatomy.* (12)

Short essay:

1. Enumerate the parts of respiratory system. Describe Broncho pulmonary segments. (7)
2. Enumerate the parts of respiratory system. Explain the anatomy of lung.(7)

Short note:

1. Diaphragm* (5)
2. Pleura** (5)
3. Structure of the nose (5)
4. Paranasal air sinuses (5)
5. Trachea(5)

Differentiate between:

1. Trachea and bronchi (4)

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UNIT-8

DIGESTIVE SYSTEM

Essays:

1. Mention the parts of the digestive system and describe the stomach under the following Headings (10)
   Relations | Peritoneal folds | Blood supply | Applied aspects
2. Draw and label parts of the gastrointestinal system. Add a note on the applied anatomy of appendix. (12)
3. Mention the different parts of digestive system. Describe the stomach under the following headings (10)
   Surfaces and borders | Relations | Blood supply | Applied anatomy
4. Explain the parts of gastrointestinal system with the help of neat labelled diagram. Add a note on esophagus (12)

Short essays:

1. Draw and label the parts of the gastrointestinal system. Describe the stomach in detail* (7)

Short notes:

1. Soft Palate (5)
2. Pancreas* (5)
3. Appendix(5)
4. Blood supply of stomach**(5)
5. Parts and relations of pancreas (5)
6. Structure of the tongue (5)
7. Large intestine (5)
8. Anal canal (5)

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9. Tongue (5)
10. Abdominal aorta (5)
11. Dorsum of tongue (5)
12. Liver(5)
13. Parts and nerve supply of tongue (5)
14. Lobes of liver (5)

Answer briefly:

1. Blood supply of stomach (4)
2. Parts and relations of pancreas (4)
3. What are the accessory organs of digestion (4)
4. List the parts of the pancreas (4)
5. Structure of tongue (4)
6. List the parts of the stomach (4)
7. Ligaments of liver(4)
8. Caecum(4)
9. Extend and constrictions of esophagus (4)

Differentiate between:

1. Small and large intestine* (4)

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UNIT-9

EXCRETORY SYSTEM

Essays:

1. Name the parts of urinary system with diagram and describe kidney under the following heading:
   External features, relations &Capsules or coverings (10)

Short notes:

1. Parts of kidney (5)
2. Structure of kidney with diagram**(5)
3. Structure of a nephron with diagram* (5)
4. Structure of urinary bladder (5)
5. Male urethra*(5)
6. Urinary bladder(5)
7. Relations of urinary bladder (5)
8. Surfaces and relations of urinary bladder(5)

Answer briefly:

1. Ureter*** (4)
2. Structure of urinary bladder*(4)
3. Structure of male urethra with diagram (4)
4. Male urethra(4)
5. Relations of right kidney**(4)
6. Male Urethra (4)

Differentiate between:

1. Urethra and ureter (4)

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UNIT-10

ENDOCRINE SYSTEM

Essays:

1. Mention the endocrine glands and describe the thyroid under the following heading: Location and relations /Blood supply /Applied aspects (10)

Short essays:

1. Enumerate the endocrine glands. Describe one of the endocrine gland in detail with clinical importance. (7)

Short notes:

1. Parts of thyroid gland (5)
2. Pancreas (5)
3. Suprarenal gland (5)
4. Parts and relations of pancreas** (5)
5. Describe the pancreas (5)
6. Thyroid gland(5)
7. Pituitary gland*(5)

Answer briefly:

1. Thymus gland*(4)
2. Parts of pituitary gland (4)
3. Structure of pancreas*(4)
4. List the parts of the adrenal gland (4)
5. Classify glands with example(4)
6. Pancreas (4)
7. Parotid gland (4)
8. Enumerate the pituitary hormones (4)

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UNIT-11

REPRODUCTIVE SYSTEM

Essays:

1. Mention the different parts of female reproductive system and describe the uterus under the following headings
   Location and parts Relations /Supports /Blood supply* (10)
2. Mention the parts of male reproductive system. Describe testis under the following headings:
   Position /coverings /blood supply /applied anatomy (10)

Short essays:

1. Name the male reproductive organs. Add a note on testes (7)
2. Draw and label parts of the female reproductive system. Describe the uterus in detail (7)

Short notes:

1. Fallopian tube (5)
2. Supports of uterus (5)
3. Mammary gland (5)
4. Coverings and applied anatomy of testis (5)
5. Structure of the uterus*(5)
6. Structure of testis (5)
7. Structure of breast*(5)
8. Prostate (5)
9. Fallopian tube*(5)
10. Coverings of testis (5)
11. Testes (5)

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Answer briefly:

1. Lobes of prostate gland (4)
2. Structure of female reproductive system*(4)
3. Prostate (4)
4. List the organs of male and female reproductive system (4)

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PHYSIOLOGY

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UNIT-1

CELL PHYSIOLOGY

Short notes:

1. Structure of Cell membrane (5)
2. Mitochondria (5)

Answer briefly:

1. Active transport (4)
2. Diffusion (2)
3. Functions of cell membrane (2)

Differentiate between:

1. Diffusion and osmosis* (4)

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UNIT-2

SKELETAL SYSTEM

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UNIT-3

MUSCULAR SYSTEM

Short essay:

1. Describe the physiological events in skeletal muscle contraction (7)

Short notes:

1. Rigor mortis (5)
2. Events at Neuromuscular junction (5)
3. Excitation-contraction coupling (5)
4. Neuromuscular junction* (5)
5. Activity at the neuromuscular junction (5)
6. Differences between monosynaptic and polysynaptic reflexes (5)
7. Neuromuscular transmission (5)
8. Differentiate between skeletal, cardiac and smooth muscles (5)
9. Action potential (5)
10. Properties of cardiac muscle (5)

Answer briefly:

1. Myasthenia gravis (2)
2. Sarcomere * (4)
3. Neurotransmitters (4)
4. Muscle proteins (2)
5. Tetany (2)
6. Resting membrane potential (2)
7. Name four muscle proteins (2)

Differentiate between:

1. Cardiac and skeletal muscle (4)
2. Skeletal muscle and smooth muscle * (4)
3. Resting membrane potential and action potential (4)

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UNIT-4

NERVOUS SYSTEM

Essay:

1. Discuss the composition, formation, circulation and functions of cerebrospinal fluid. (10)

Short essays:

1. Define reflex. Explain reflex arc with the help of a diagram. Explain stretch reflex (1+3+3=7)
2. Describe the spinothalamic and pyramidal tract pathways. (3+4=7)
3. Define synapse. What are the properties of synapse (7)

Short notes:

1. Cerebrospinal fluid. (5)
2. Functions of hypothalamus (5)
3. Pain pathway (5)
4. Functions of thalamus * (5)
5. Neuron (5)
6. Parkinson's disease (5)
7. Composition and functions of cerebrospinal fluid. (5)
8. Pyramidal tracts (5)
9. Properties of synapse (5)
10. Types of neuroglia (5)

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Answer briefly:

1. Functions of cerebellum (2)
2. Referred pain (2)
3. Neuron (2)
4. Reflex arc (2)
5. Name the ascending tracts (2)
6. Components of Basal ganglia (2)
7. Referred pain * (2)
8. Two types of neuroglia (2)
9. Functions of hypothalamus (2)
10. Define hemiplegia (2)
11. Parkinsonism (2)
12. Two ascending tracts and the sensations carried by them (2)
13. What are the functions of CSF (4)

Differentiate between:

1. Actions of sympathetic and parasympathetic nervous system (4)

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UNIT-5

PHYSIOLOGY OF BLOOD

Essays:

1. Discuss the blood groups in detail. Add a note on blood transfusion reactions. (7+3=10)
2. Define erythropoiesis. Describe the different stages of erythropoiesis. Add a note on Anemia (1+6+3=10)
3. Explain the steps in erythropoiesis. Describe the various factors influencing Erythropoiesis. Add a note on functions of RBCs. (4+4+4=12)
4. Define hemostasis and mention the steps in hemostasis. Describe the intrinsic and extrinsic coagulation pathways. Add a note on anti-coagulants. (4+4+4=12)
5. Describe the stages of erythropoiesis. What are the factors regulating erythropoiesis. Add a note on iron deficiency anemia (4+4+2=10)
6. Describe in detail the mechanism of coagulation. Add a note on hemophilia (8+2)
7. Define erythropoiesis. Describe the different stages of erythropoiesis. Add a note on factors regulating it. (1+5+4=10)
8. Discuss the blood groups in detail. Add a note on blood transfusion reactions (5+5=10)
9. Enumerate the coagulation factors. Explain the intrinsic pathway of coagulation. Add a note on hemophilia (2+6+2=10)

Short essays:

1. Describe the different types of blood groups. Add a note on Rh incompatibility (4+3=7)
2. What are the different types of blood groups. Briefly explain the functions of leukocytes. (3+4)

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Short notes:

1. ABO blood group systems (5)
2. Structure and functions of platelets * (5)
3. Erythropoiesis (5)
4. Functions of WBC (5)
5. Describe the eosinophil with the aid of a diagram (5)
6. Haemostasis (5)
7. Types and functions of immunoglobulins (5)
8. Cell mediated immunity (5)
9. Functions of plasma proteins (5)
10. Classify Anemia with examples (5)
11. Blood coagulation (5)

Answer briefly:

1. ESR * (2)
2. Neutrophils (2)
3. Hemophilia (2)
4. Lymphocytes (2)
5. Jaundice (2)
6. Landsteiner's law (2)
7. Functions of plasma proteins (4)
8. Sickle cell anemia (2)
9. Intrinsic factor (2)
10. Two functions of hemoglobin (2)
11. Cross matching (2)
12. Anti-coagulants (2)
13. Pernicious anemia(2)
14. Functions of Hemoglobin (2)
15. Rh incompatibility (2)
16. Leukemia (2)
17. Clotting time (2)
18. Neutrophil (2)
19. Mention any two tests for anemia (2)

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Differentiate between:

1. Neutrophil and lymphocyte (4)
2. Extracellular fluid and intracellular fluid (4)
3. Innate and acquired immunity (4)
4. Homeostasis and hemostasis (4)
5. Rh incompatibility and ABO incompatibility (4)
6. Erythropoiesis and erythrolysis (4)

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UNIT-6

CARDIOVASCULAR SYSTEM

Essay:

1. Define cardiac cycle. Discuss the different phases of cardiac cycle. Add a note on heart sounds. * (2+8+2=12)
2. Define arterial blood pressure and give its normal value. What is hypertension? Explain the long term regulation of blood pressure * (3+3+4=10)
3. Define cardiac output. Discuss the factors regulating stroke volume. What is cardiac index? (2+6+2=10)

Short essays:

1. Define cardiac cycle and mention the duration. Describe the various events in the Cardiac cycle * (3+4=7)
2. Define blood pressure and mention its normal values. Describe the various factors affecting blood pressure (3+4=7)
3. Define cardiac output and mention the normal value. Describe the factors influencing cardiac output. (3+7=7)

Short notes:

1. Normal ECG (5)
2. Describe the waves of ECG and mention its uses. (5)
3. Conduction system of the heart (5)
4. Coronary circulation (5)
5. Circulatory shock (5)
6. Factors affecting blood pressure (5)

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7. Factors influencing cardiac output (5)
8. Short term regulation of blood pressure (5)
9. Second heart sound (5)

Differentiate between:

1. First and second heart sound (4)

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UNIT-7

THE RESPIRATORY SYSTEM

Essays:

1. Mention the normal respiratory rate. Explain the oxygen transport in the blood. Add a note on hypoxia (1+5+4=10)
2. Explain the various lung volumes and capacities with the help of a diagram. Add a note on vital capacity. (4+4+2=10)
3. Describe the transport of oxygen and carbon-dioxide in the blood. Add a note on Oxygen Dissociation Curve (4+4+4=12)
4. Mention the normal respiratory rate. What is surfactant and enumerate its functions.
   Describe lung volumes with the aid of a diagram (1+4+5=10)
5. Define hypoxia and name the types. Describe anemic hypoxia (1+2+7=10)
6. Mention modes of oxygen transport. Draw a labeled diagram of oxygen dissociation curve*. (5+5=10)
7. Explain the regulation of respiration. Define the lung volumes and capacities with their normal values (12)

Short essays:

1. Draw the normal oxygen – hemoglobin dissociation curve (ODC). What are the factors causing shift of ODC (2+5=7)
2. Discuss the modes of carbon dioxide transport in blood. Add a note on asphyxia (8+2=10)

Short notes:

1. Non respiratory functions of lung (5)
2. Oxygen transport (5)

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3. Oxygen dissociation curve (5)
4. Describe the mechanism of inspiration and expiration (5)
5. Chemical regulation of respiration. (5)
6. Neural regulation of respiration 95)
7. Carbon-dioxide (CO2) transport * (5)
8. Surfactant (5)
9. Lung volumes and capacities (5)
10. Functions of lungs (5)
11. Respiratory volumes (5)

Answer briefly:

1. Bohr effect * (2)
2. Cyanosis * (2)
3. Artificial respiration (2)
4. Dyspnoea (2)
5. Tidal volume * (2)
6. Define residual volume and vital capacity (2)

TCON ACADEMIC CLUB

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UNIT-8

THE DIGESTIVE SYSTEM

Essays:

1. Discuss the composition and functions of gastric juice. Add a note on deglutition. Define jaundice (6+2+2=10)

Short notes:

1. Functions of liver (5)
2. Functions of bile (5)
3. Deglutition (5)
4. Stages of deglutition (5)
5. Composition and functions of pancreatic juice (5)
6. Phases of gastric secretion (5)
7. Functions of the large Intestine (5)
8. Movements of small intestine (5)
9. Functions of gastric juice (5)

Answer briefly:

1. Movements of small intestine (2)
2. Functions of colon. (2)
3. Functions of gall bladder (2)
4. Functions of saliva (2)
5. Composition of saliva (2)
6. Gastric juice (2)
7. Peristalsis (2)
8. Bile salts (2)
9. Functions of stomach (2)
10. Functions of gastric acid (2)

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TCON ACADEMIC CLUB

UNIT-9

RENAL SYSTEM

Essays:

1. Define glomerular filtration rate and give its normal value. What are the factors regulating GFR. Add a note on osmotic dieresis *(2+6+2=10)
2. Describe the various events that occur in proximal convoluted tubule. What is the normal urine output (8+2=10)
3. Mention the normal urine output per day. Discuss the three steps in the formation of urine. Add a note on re-absorption of water by the nephron (1+6+3=10)
4. Define glomerular filtration rate. Give its normal value. Explain in detail the factors influencing glomerular filtration rate. Add a note on glucose reabsorption by the kidney * (1+1+6+4=12)
5. Define glomerular filtration rate and mention its normal value. Discuss the factors affecting GFR. Add a note on effective filtration pressure (2+6+2=10)

Short essays:

1. Define glomerular filtration rate (GFR). What are the factors influencing it (2+5=7)

Short notes:

1. Juxtaglomerular apparatus * (5)
2. Formation of urine (5)
3. Factors affecting glomerular filtration rate * (5)
4. Glomerular filtration (5)
5. Mechanism of urine formation (5)
6. Loop of Henle (5)
7. Function of nephron (5)

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Answer briefly:

1. Draw and label a nephron (2)
2. Micturition reflex (2)
3. Renal threshold (2)
4. Functions of Kidney (2)
5. Juxta glomerular apparatus * (2)
6. Functions of Saliva (2)
7. Dialysis (2)
8. List two components of counter current mechanism (2)
9. Mechanism of urine formation (4)

Differentiate between:

1. Counter-current exchanger and counter-current multiplier (4)

TCON ACADEMIC CLUB

UNIT-10

SKIN & TEMPERATURE REGULATION

Short notes:

1. Functions of skin (5)

Answer briefly:

1. Role of skin in regulation of body temperature (2)
2. Hypothermia (2)

Asterisks (*) denotes repetition.

UNIT-11

SPECIAL SENSES

Short notes:

1. Myxedema (5)
2. Visual pathway (5)
3. Refractive errors of eye* (5)
4. Mechanism of hearing (5)
5. Contents and functions of middle ear (5)

Answer briefly:

1. Myopia (2)
2. Functions of middle ear (2)
3. Presbyopia (2)
4. Organ of corti (2)
5. Primary taste sensations (2)
6. Color blindness (4)
7. Refractory period (2)
8. Aqueous humor (2)
9. Astigmatism (2)
10. Visual receptors (2)
11. Anosmia (2)
12. Hypermetropia (2)

Differentiate between:

1. Rods and cones (4)

TCON ACADEMIC CLUB

Asterisks (*) denotes repetition.

UNIT-12

ENDOCRINE SYSTEM

Essays:

1. How is thyroxin synthesized. Explain the physiological actions of thyroxin. What is myxedema (3+5+2=10)
2. What are the actions of thyroid hormone. What is cretinism (8+2=10)
3. Mention the hormones secreted by the anterior pituitary gland. Explain the actions of growth hormone. Mention one deficiency disease of this hormone (3+5+2=10)

Short notes:

1. Hormones acting on stomach (5)
2. Describe the main clinical features of diabetes mellitus (5)
3. Enzymes of Pancreas and their function (5)
4. Functions of insulin (5)
5. Anti-diuretic hormone (5)
6. Enumerate the hormones of anterior pituitary gland (5)
7. Functions of thyroid hormone (5)
8. Hormones of posterior pituitary and their functions * (5)
9. Functions of growth hormone (5)
10. Hormones regulating blood calcium (5)
11. Functions of Thyroid Gland (5)

Answer briefly:

1. Acromegaly * (2)
2. Cretinism (2)
3. Hormones regulating pancreatic juice secretion (4)
4. Neuro-endocrine reflex (4)

Asterisks (*) denotes repetition.

5. Hormones of pancreas and their functions (2)
6. Impedance matching in the ear (2)

Differentiate between:

1. Actions of insulin and glucagon (4)
2. Parathyroid hormone and calcitonin (4)
3. Enzymes and hormones (4)

TCON ACADEMIC CLUB

Asterisks (*) denotes repetition.

UNIT-13

REPRODUCTIVE SYSTEM

Essays:

1. Describe the ovarian and uterine changes occurring during a normal menstrual cycle. Mention the important hormones involved in each phase. What is LH surge(4+4+2=10)
2. What is menstrual cycle. Describe the different stages of menstrual cycle. Explain the role of hormones involved in it (1+6+3=10)
3. Describe menstrual cycle and its hormonal basis (5+5=10)
4. Mention the normal duration of menstrual cycle. Describe the phases of menstrual cycle. Discuss the importance of ovulation time.(1+4+5=10)

Short essays:

1. Describe the uterine changes during the various phases of the menstrual cycle. Add a note on contraceptive methods (4+3=7)
2. What is menstrual cycle. Briefly explain the changes during menstrual cycle.(7)

Short notes:

1. Menstrual cycle (5)
2. Functions of placenta * (5)
3. Pregnancy tests * (5)
4. Stages of spermatogenesis * (5)
5. Functions of testes (5)
6. Tests of ovulation (5)
7. Milk ejection reflex (5)
8. Female contraception (5)
9. Enumerate the ovarian hormones. Describe any one in detail (5)

Asterisks (*) denotes repetition.

Answer briefly:

1. Copper-T (2)
2. Ovulation (2)
3. Functions of testosterone (2)
4. Oral contraceptive pills in the female (2)
5. Spermatogenesis (2)
6. Corpus luteum * (4)
7. Functions of oxytocin * (2)
8. Basis for pregnancy tests (2)
9. Oral pill (2)
10. Define menarche and menopause (2)
11. Hormones for lactation (2)
12. Grafian follicle (2)
13. Functions of ovary (2)
14. Intrauterine contraceptive devices (2)

Differentiate between:

1. Proliferative phase and secretory phase of menstrual cycle (4)\

TCON ACADEMIC CLUB

UNIT 14

LYMPHATIC & IMMUNOLOGICAL SYSTEM

Essay

1. What is cellular immunity? Describe the mechanism of cellular immunity. Add a note on acquired immune deficiency syndrome (2+4+4=10)

Differentiate between

1. Active and passive immunity (4)

Asterisks (*) denotes repetition.

MICROBIOLOGY

TCON ACADEMIC CLUB

Asterisks (*) denotes repetition.

UNIT-1

INTRODUCTION

Answer briefly:

1. Name four contributions of Louis Pasteur (2)
2. Pasteurization (2)
3. Louis Pasteur*(2)
4. Koch's postulates (2)

TCON ACADEMIC CLUB

Asterisks (*) denotes repetition.

UNIT-2

GENERAL CHARACTERISTICS OF MICROBES

Short notes:

1. Virus cultivation(5)
2. Bacterial toxins(5)
3. Bacterial staining techniques(5)
4. Bacterial growth curve(5)
5. Carriers(5)

Answer briefly**:**

1. Bacterial exotoxins (2)
2. Anaerobic culture media(2)
3. Acid fast staining(2)
4. Name two anaerobic bacteria and two anaerobic culture methods (2)
5. Drugs acting on the cell wall of bacteria(2)
6. Transport medium(2)
7. Name two anaerobic culture methods (2)
8. Transport media (2)
9. Bacterial growth curve(2)
10. Acid fast staining(2)
11. Negri bodies(2)
12. What are selective media. Give two examples(2)
13. Bacterial filters(2)
14. Transport medium(2)
15. Inspissation(2)
16. Enumerate transport media for specimens. Enumerate transport media for specimen (2)
17. Gram staining(2)

Asterisks (*) denotes repetition.

18. Enriched medium*(2)
19. Tyndallisation (2)
20. Specimen collection in meningitis (2)
21. Negative staining (2)
22. Bacterial flagella (2)
23. Selective medium*(2)
24. Gram stain picture of staphylococcus aureus (2)
25. Sample collection in urinary tract infection (2)
26. Bacterial spores (4)

Differentiate between:

1. Flagella and fimbriae (4)
2. Gram positive and gram negative cell wall (4)
3. Exotoxins and endotoxins (4)
4. Gram positive and gram negative cell wall (4)

TCON ACADEMIC CLUB

Asterisks (*) denotes repetition.

UNIT-3

INFECTION CONTROL

Essays:

1. Define biomedical waste and name the different types of hospital wastes. Discuss in detail the methods of disposal of hospital waste with special reference to biomedical waste. (2+1+6+3)
2. Define disinfection and name various types of disinfectants. Discuss the role of halogens in chemical disinfection and add a note in the disinfection of operation theatre. (1+2+6+3)
3. List the bacteria causing urinary tract infections. Describe in detail the sample collection in a one year old baby, an adult male, female and a catheterized patient. Briefly describe the transport, processing and culture of a urine sample. (3+5+4)
4. Define Health care associated infections. Describe the various routes of transmission, prevention and control of health care associated infections. Enumerate the role of nurse in hospital infection control (2+7+3)
5. Define and classify sterilization. Explain the principle, functions and uses of autoclave. (1+2+1+4+2=10)
6. Define Sterilization. Classify the various methods of ste1ilization. Discuss in detail methods of moist heat sterilization (1+2+7=10)
7. Define sterilization. Enumerate different methods of sterilization. Describe in detail the working of autoclave. Mention the sterilization control for autoclave. (2+2+4+2 =10)
8. Define hospital acquired infections (HAI). Describe the various sources, routes of transmission and common types of HAI. Add a note on standard precaution. (1+2+2+2+3 =10)
9. Define sterilization and classify sterilizing agents. Explain in detail sterilization by moist heat. Enumerate the role of a nurse in sterilization.(12).

Asterisks (*) denotes repetition.

Short essays:

1. Classify disinfectants and name the chemical agents used for skin antisepsis. Describe the steps of performing hand hygiene. (2+2+3)
2. Bio medical waste management principles and steps. (3+4)
3. Define hospital acquired infections (HAI). Describe the various sources, routes of transmission and the common types of HAI (7)

Short notes:

1. Autoclave (7)
2. Hospital infection control committee.(7)
3. Different sources of infection.(7)
4. Chemical disinfectants*(7)
5. Standard precautions. (7)
6. Nosocomial infections** (7)
7. Modes of transmission of infection.(7)
8. Dry heat methods of sterilization.(7)
9. Biomedical waste management***(7)
10. Role of central sterile supplies department in a hospital.(7)
11. Hot air oven**(7)
12. Autoclave**(7)
13. Barrier nursing*(7)
14. Operating theatre sterilization (7)
15. Role of nurses in hospital infection control committee (7)
16. Universal precautions (7)
17. Standard safety measures (7)
18. Hand hygiene (7)
19. Universal precautions (5)

TCON ACADEMIC CLUB

Asterisks (*) denotes repetition.

Answer briefly:

1. Carriers. (2)
2. Name four universal precaution methods (2)
3. Gaseous disinfectants (2)
4. Fumigation (2)
5. Post exposure prophylaxis after a needle stick injury. (2)
6. Disposal of sharps (2)
7. Prevention of surgical site infections. (2)
8. Personal protective equipment (2)
9. Droplet precaution (2)
10. Define biological vectors and give two examples (2)
11. Definitive host (2)

Differentiate between:

1. Agglutination and precipitation (4)

TCON ACADEMIC CLUB

Asterisks (*) denotes repetition.

UNIT-4
PATHOGENIC ORGANISMS

• BACTERIOLOGY

Essays:

1. Describe the pathogenesis, laboratory diagnosis and prophylaxis of diphtheria (10)
2. Describe the pathogenesis, laboratory diagnosis and prophylaxis of typhoid fever.(3+4+3)
3. Describe the source of infection, mode of transmission, pathogenesis and laboratory diagnosis of pulmonary tuberculosis (1+1+4+4=10)
4. Describe the pathogenesis, laboratory diagnosis and prophylaxis of tetanus. (3+3+4)
5. Describe the morphology, pathogenesis, clinical conditions, lab diagnosis and treatment of staphylococcus aureus. (1+2+2+3+2)
6. Name four bacteria causing food-borne infections. Describe the morphology, clinical features and laboratory diagnosis of cholera. (2+1+2+5= 10)
7. Name two bacteria causing enteric fever. Describe the mode of transmission, laboratory diagnosis and prophylaxis of enteric fever (1+1+4+4)
8. Name the bacteria causing enteric fever. Describe the route of transmission, clinical features, laboratory diagnosis and prophylaxis of enteric fever. (1+2+2+3+2=10)
9. Name the causative agent of cholera. Describe the pathogenesis, clinical features, laboratory diagnosis and prophylaxis of cholera. (1+3+1+3+2)

Asterisks (*) denotes repetition.

10. Name the organism causing tetanus. Describe in detail about the pathogenesis, laboratory diagnosis and prophylaxis of tetanus. (1+3+3+3=10)
11. Classify mycobacteria. Describe in detail the pathogenesis, laboratory diagnosis and prophylaxis of tuberculosis (2+3+3+2 = 10)
12. Name the causative agent of tetanus. Describe the pathogenesis, clinical features and prophylaxis of tetanus. (1+3+2+4)
13. Define Urinary tract infection(UTI). List the organisms commonly causing UTI. What is significant bacteriuria. Describe the various methods of sample collection and lab diagnosis. (1+2+1+4+2=10)

Short essays:

1. Describe the clinical features and laboratory diagnosis of enteric fever (2+5)
2. Describe the pathogenesis and laboratory diagnosis of pulmonary tuberculosis* (2+5)
3. Laboratory diagnosis of cholera and measures to be taken to prevent spread of infection in a cholera outbreak. (4+3)
4. Describe the infections caused by staphylococcus aureus. Expand MRSA. Add a note on the measures to prevent spread of MRSA infections in hospitals. (3+1+3)
5. Describe the sample collection and laboratory diagnosis of pulmonary tuberculosis.Expand the terms RNTCP and DOTS (2+3+1+1)
6. Describe the morphology, clinical features and laboratory diagnosis of cholera. (7)

Short notes:

1. Laboratory diagnosis of cholera* (7)
2. Standard tests for syphilis. (7)
3. Urinary tract infection (7)
4. Catheter associated urinary tract infection (7)

Asterisks (*) denotes repetition.

5. Swine flu (7)
6. Methicillin resistant staphylococcus aureus (7)
7. Immunoprophylaxis of tetanus (7)
8. Collection and transport of specimen in cholera (7)
9. Laboratory diagnosis and prevention of swine flu (H1N1) infection (7).
10. Laboratory diagnosis of urinary tract infection* (7)
11. Leptospirosis (7)
12. Prophylaxis of tetanus** (7)
13. Anthrax (7)
14. E.coli (7)
15. Laboratory diagnosis of pulmonary tuberculosis** (7)
16. Staphylococcus aureus (7)
17. Lab diagnosis of syphilis (7)
18. Bacillary dysentery** (7)
19. Morphology and Pathogenesis of vibrio cholera (7)
20. Laboratory diagnosis of gas gangrene (7)
21. Meningococcal meningitis (7)
22. Plague (7)
23. Pneumococcal pneumonia (7)
24. Morphology and lab diagnosis of neisseriagonorrhoeae (7)
25. Life cycle of echinococcusgranulosus (7)
26. Leptospirosis* (7)
27. Lab diagnosis of meningitis (7)
28. Lab diagnosis of pulmonary tuberculosis (5)

Answer briefly:

1. Streptococcal infections (2)
2. Mantoux test * (2)
3. List four staphylococcal infections. (2)
4. Botulism (2)
5. Name four gram negative bacilli (2)
6. Name four zoonotic diseases (2)

Asterisks (*) denotes repetition.

7. Methicillin resistant staphylococcus aureus (2)
8. Laboratory diagnosis of gonorrhea. (2)
9. Laboratory diagnosis of leptospirosis (2)
10. Congenital syphilis (4)
11. Prophylaxis of diphtheria (4)
12. Diphtheria (4)
13. Gonorrhoea (4)
14. Widal test
15. Toxic shock syndrome (2)
16. Morphology of *neisseria meningitides*(2)
17. Name four bacteria causing food poisoning (2)
18. Enumerate any four zoonotic infections (2)
19. Hydatid cyst*(2)
20. Name four bacteria causing pyogenic meningitis (2)
21. What is S.T.D. Give two examples of bacterial S.T.D. (2)
22. Name two multidrug resistant organisms (2)
23. Name four gram positive bacteria (2)
24. VDRL(2)
25. Infections caused by E.coli(2)
26. Name four common infections caused by staphylococcus aureus(2)
27. NIH swab* (2)
28. Name four bacteria causing food poisoning (2)
29. Toxic shock syndrome (2)
30. Non-gonococcal urethritis (2)
31. Specimen collection in meningitis (2)
32. Swine flu (2)
33. Prophylaxis of diphtheria (2)
34. Anthrax (2)
35. Gram stain picture of staphylococcus aureus (2)

Differentiate:

1. Exotoxins and endotoxins*(4)

Asterisks (*) denotes repetition.

- **VIROLOGY**

Essays:

1. Name the methods of transmission of HIV. Describe the pathogenesis, laboratory diagnosis and prophylaxis of HIV infection. (2+2+4+2)
2. Describe the morphology of rabies virus. Describe the pathogenesis, laboratory diagnosis and prophylaxis of rabies. (2+2+3+3)
3. Enumerate the arbo-viral infections in India. Discuss the pathogenesis, complications, laboratory diagnosis and prophylaxis of dengue fever. (2+2+2+2+2)
4. Name the blood borne micro-organisms. List the blood borne virus infections. Describe the morphology of hepatitis B, modes of spread, laboratory diagnosis and prevention of hepatitis B infection. (2+2+8)
5. Describe the morphology, clinical features and laboratory diagnosis of HIV/AIDS. Add a note on post exposure prophylaxis. (1+2+6+3)
6. Describe the morphology, routes of infection, laboratory diagnosis and prophylaxis of hepatitis B virus. (1+2+4+3=10)
7. Explain the morphology, pathogenesis, mode of infection, clinical symptoms and lab diagnosis of HIV-AIDS (2+2+2+2+2=10)
8. List the hepatitis viruses. Explain the structure, pathogenicity. lab diagnosis and prophylaxis of hepatitis B virus (2+1+4+2+1=10)
9. Name the causative agent of polio. Describe briefly the route of transmission, clinical features, lab diagnosis and prophylaxis of polio. (1+1+2+3+3 =10)
10. List out the viruses causing hepatitis. Describe in detail the pathogenesis, laboratory diagnosis and prophylaxis of hepatitis B (2+3+3+2 =10)
11. Describe the structure, mode of infection, pathogenesis, clinical features and lab diagnosis of human immunodeficiency virus. (2+1+2+1+4)

Asterisks (*) denotes repetition.

Short essays:

1. Discuss the morphology, modes of transmission and laboratory diagnosis of hepatitis B virus (1+2+4)
2. Describe the morphology, modes of transmission and laboratory diagnosis of HIV/AIDS.(1+2+4)

Short notes:

1. Laboratory diagnosis and prophylaxis of poliomyelitis(7)
2. Prophylaxis of rabies * (7)
3. Serological markers in hepatitis B (7)
4. Dengue fever and its complications (7)
5. Opportunistic infections in AIDS (7)
6. Laboratory diagnosis of dengue (7)
7. Laboratory diagnosis and prevention of hepatitis B infection (7)
8. General and specific measures to prevent hepatitis B virus infection (7).
9. Morphology and pathogenesis of HIV (7)
10. Rubella (7)
11. Laboratory diagnosis of HIV (7).
12. Measles (7)
13. Dengue fever (7)
14. Prophylaxis of poliomyelitis (7)
15. Immuno-prophylaxis of poliomyelitis (7)

Answer briefly:

1. Japanese encephalitis.(2)
2. Kyasanur Forest Disease(2)
3. Name four zoonotic diseases(2)
4. Post exposure prophylaxis of rabies(2)
5. Dengue Fever* (2)
6. Enumerate any four zoonotic infections(2)
7. Enumerate any four arbo-viral diseases prevalent in India** (2)

Asterisks (*) denotes repetition.

8. Immunoprophylaxis of hepatitis B virus (2)
9. Common modes of transmission of HIV(2)
10. Types of Hepatitis Viruses (2)
11. Name four viruses transmitted through blood transfusion. (2)
12. Structure of HIV (2)
13. Rubella (2)
14. Name four agents causing diarrhea in AIDS patients (2)
15. Prophylaxis of rabies (4)

Differentiate between**:**

1. Oral polio vaccine and injectable polio vaccine(OPV and IPV)**(4)

TCON ACADEMIC CLUB

- **MYCOLOGY**

Short notes:

1. Candidiasis ****** (5)
2. Dermatophytosis.*** (5)
3. Life cycle of taenia solium and its laboratory diagnosis (5)
4. Cryptococcosis. ** (5)
5. Candida albicans (5)
6. Superficial mycoses * (5)

Answer briefly:

1. Dimorphic fungi * (2)
2. Cryptococcal meningitis (2)
3. Dermatophytes ** (4)
4. Cryptococcus neoformans (4)
5. Opportunistic fungal infections (4)

Asterisks (*) denotes repetition.

6. Name four opportunistic fungal infections (2)
7. Candida albicans (2)
8. Name four opportunistic fungal infections in man (2)
9. Cryptococcosis (2)
10. Name four dimorphic fungi (2)

- **PARASITOLOGY**

Essays:

1. Describe the life cycle and clinical features of plasmodium falciparum. Describe briefly about lab diagnosis of malaria * (5+2+3=10)
2. Name the different species of malarial parasite. Describe the life cycle, clinical features laboratory diagnosis and complications of plasmodium falciparum malaria. (2+2+2+2+2=1 0)
3. Explain morphology, pathogenesis and lab diagnosis of plasmodium vivax. (2+3+5=10)

Short essays:

1. Describe the clinical features and laboratory diagnosis of falciparum malaria (2+5=7)
2. Describe the clinical features and laboratory diagnosis of intestinal amoebiasis (2+5=7)

Short notes:

1. Life cycle of wuchereria bancrofti. (5)
2. Life cycle of plasmodium vivax (5)
3. Entamoeba histolytica * (5)
4. Enterobius vermicularis (5)

Asterisks (*) denotes repetition.

5. Hook worm infections (5)
6. Microfilaria **** (5)
7. Laboratory diagnosis of malaria. * (5)
8. Amoebiasis ** (5)
9. Lab diagnosis of enteric fever * (5)
10. Life cycle of ascaris lumbricoides (5)
11. Giardiasis * (5)
12. Life cycle of hookworm (5)

Answer briefly:

1. Trichomonas vaginalis (2)
2. Eggs of round worms (2)
3. Giardia lamblia (2)
4. Microfilaria (2)
5. Trichomonas vaginalis (2)
6. Name four intestinal nematodes (2)
7. Erythrocytic shizogony of plasmodium falciparum (2)
8. Tinea solium (2)
9. Tinea versicolor * (2)
10. Falciparum malaria (4)
11. Trophozoite of entamoeba histolytica (2)
12. Name four common species causing malaria (2)
13. Name four vector born infections (5)
14. Trichomonas vaginalis (5)
15. Name the definitive and intermediate hosts of filarial worm (2)
16. Name the definitive and intermediate hosts of *taenia saginata (2)*

Difference between:

1. Plasmodium vivax and plasmodium falciparum in peripheral smear (4)

Asterisks (*) denotes repetition.

UNIT-5

IMMUNITY

Essays:

1. Define and classify hypersensitivity. Explain the type I hypersensitivity in detail(1 + 2 + 7 =10)
2. Define and classify hypersensitivity. Explain type 1 hypersensitivity.(1+2+7=10)
3. Define immunity. Describe the factors affecting innate immunity.(2+8)
4. Define and classify hypersensitivity reactions. Describe the mechanism and presentations of type I hypersensitivity reaction in detail(1+2+7)

Short essays:

1. Type I hypersensitivity reaction* (5)
2. Type III hypersensitivity reaction. (5)
3. ELISA (5)
4. Type IV hypersensitivity reaction (5)
5. Anaphylaxis (5)
6. Describe the principle and application of agglutination (5)
7. DPT vaccine (5)
8. Delayed hypersensitivity reaction* (5)
9. Active immunity (5)
10. Innate immunity* (5)
11. Passive immunity (5)
12. Antibiotic sensitivity testing (5)
13. Killed vaccines and live vaccines (5)
14. Acquired immunity (5)
15. National immunization programme (5)
16. National immunization schedule*(5)

Asterisks (*) denotes repetition.

Answer briefly:

1. Live viral vaccines(2)
2. Passive immunity*(2)
3. Active immunity (2)
4. DPT vaccine*(2)
5. Pulse polio immunization (2)
6. BCG vaccine (2)
7. MMR vaccine (4)
8. Type I hypersensitivity (4)
9. VDRL/RPR test (4)
10. Widal test (2)
11. IgM antibody (2)
12. BCG(2)
13. Herd immunity*(2)
14. MMR vaccine*(2)
15. IgG (2)
16. Classify hypersensitivity reactions (2)
17. T-lymphocytes.(2)
18. Name four viral vaccines (2)
19. IgA (2)
20. Immunisation schedule* (2)
21. IgM (2)
22. Name four bacterial vaccines (2)
23. Classify active immunity with examples (2)
24. IgG antibody (2)

Differentiate between:

1. Active and passive immunity (4)
2. Immunoglobulin M (IgM) and Immunoglobulin G (IgG) (4)
3. Live and killed vaccines* (4)
4. T cell and B cell (4)
5. Type I and type IV hypersensitivity reactions (4)
6. Immediate and delayed type of hypersensitivity (4)

Asterisks (*) denotes repetition.

TCON ACADEMIC CLUB

NUTRITION

TCON ACADEMIC CLUB

Asterisks (*) denotes repetition.

UNIT-1

INTRODUCTION

Short essays:

1. Explain the role of nutrition in maintaining health (10)
2. Explain national nutrition policy with its specific goal (10)
3. Write methods of nutritional assessment and role of nurse in nutrition education *(7)

Short notes:

1. Factors affecting food and nutrition*(5)
2. Classification of food (5)
3. Nutritional problems in India (4)

Answer briefly:

1. Elements of Nutrition(3)
2. Classification of foods*(3)
3. Basic concepts of nutrition(3)
4. Factors affecting food and nutritional intake(3)
5. Calorific value of nutrients(3)

Asterisks (*) denotes repetition.

UNIT-2

CARBOHYDRATE

Short essays:

1. Explain the digestion and absorption of carbohydrates. List the functions and Characteristics of carbohydrates* (7)
2. Write an essay on TCA Cycle. Add a note on its energetic (10)
3. Explain the absorption, digestion and metabolism of carbohydrates (7)

Short notes:

1. Classifications of carbohydrates (5)
2. Digestion and absorption of carbohydrates (5)
3. Role of fiber* (4)
4. Digestion of carbohydrates (4)
5. Functions of carbohydrates* (4)
6. Daily requirement and functions of fibre (5)

Answer Briefly

1. Glycemic index (4)

UNIT-3

FATS

Short essay:

1. Explain the classification, functions, requirements and sources of fat (10)

Short note:

1. Essential fatty acids (4)
2. Properties of fats*. (5)
3. Classification of fats (5)
4. Causes and prevention of obesity (5)

Asterisks (*) denotes repetition.

UNIT-4

PROTEIN

Short essays:

1. Explain protein under the following headings: (10)
   -Digestion / Sources / Functions / Deficiencies
2. Discuss about protein energy malnutrition in detail* (7)
3. Explain the classification, digestion and deficiency manifestations of protein. (7)

Short notes:

1. Classification of proteins (5)
2. Sources and functions of proteins. (5)
3. Dietary sources of proteins. (4)

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UNIT-5

ENERGY

Short essays:

1. Define basal metabolic rate. Explain its determination and factors affecting it. (10)

Short notes:

1. Define and classify BMI (5)
2. What are the factors which affecting BMR* (5)

Asterisks (*) denotes repetition.

UNIT-6

VITAMINS

Short essays:

1. Explain about water soluble vitamins in detail (7)
2. Explain vitamin A under the following heads: (10)
   - Sources / Functions / RDA / Deficiencies

Short note:

1. Sources, functions and deficiency of vitamin A. (5)
2. Deficiency symptoms of thiamine (5)
3. Vitamin D deficiency diseases (5)
4. Scurvy*(4)
5. Pellagra*(4)
6. Vitamin D deficiency (5)
7. Metabolism of vitamin D (5)
8. Functions, sources and requirement of vitamin B_{12} (5)

Answer briefly

1. Rickets (4)

Asterisks (*) denotes repetition.

UNIT-7

MINERALS

Short essays:

1. Describe the requirements, sources and deficiency of dietary iron (7)
2. What is the requirement of calcium in normal diet? Mention the food sources, factors affecting absorption and deficiency manifestation of calcium (7)

Short notes:

1. Functions of calcium (5)
2. Sources of calcium (5)
3. Goiter (5)
4. Calcium deficiency*(4)
5. Causes and prevention of fluorosis (5)

Answer briefly:

1. Osteomalacia (3)

Asterisks (*) denotes repetition.

UNIT-8

WATER

Short essay:

1. Explain the functions, requirements and management of water (10)

Short notes:

1. Regulation of water metabolism (5)
2. Importance of water in the body (5)
3. Dietary methods used to assess the nutritional status (5)
4. Maintenance of fluid and electrolyte balance (5)

Answer briefly:

1. Water deprivation (3)

TCON ACADEMIC CLUB

Asterisks (*) denotes repetition.

UNIT-9

COOKERY RULES AND PRESERVATION OF NUTRIENTS

Short essays:

1. Differentiate between food adulteration and food additives (10)

Short note:

1. Methods of food preservation** (5)
2. Methods of cooking* (5)
3. Household methods of food preservation (5)
4. Prevention of food adulteration act (5)
5. Canning (4)
6. Dry heat methods of cooking (4)
7. Principles of cooking and serving* (4)
8. Domestic methods of food preservation. (4)
9. Methods of preservation of nutrients (4)

Asterisks (*) denotes repetition.

UNIT-10

BALANCED DIET

Short essays:

1. Define balanced diet. Mention the principles involved in preparing a low cost nutritious diet. Prepare a day's menu for an adolescent girl. *(10)
2. What is balanced diet? Narrate a day's menu for a normal pregnant woman in second trimester. *(7)
3. List down the principles of meal planning and plan a day's menu for an expectant woman. (10)

Short notes:

1. Prepare a day's menu of balanced diet for an adolescent girl. (5)
2. Prescribe a balanced diet for a preschool child (4)
3. Principles of Menu Planning. *(4)
4. Plan and discuss a diet for an adult sedentary women suffering from diabetes mellitus (5)
5. Plan a diabetic diet for a 60 years old woman with 164 cms tall and 72 kg weight (4)

Answer briefly:

1. Plan a day's menu for an adult woman doing sedentary work (3)
2. Menu plan for a sedentary diabetic patient* (4)
3. Naturopathy diet * (3)
4. Types of menus (3)
5. Balance diet. (4)

Asterisks (*) denotes repetition.

TCON ACADEMIC CLUB

UNIT-11

ROLE OF NURSE IN NUTRITIONAL PROGRAMMES

Short essays:

1. Explain the nutritional problems in India *(7)

Short notes:

1. Mid-day meal programme (5)
2. Measures taken to prevent anemia in India (5)
3. Nutritional programme in India (4)

Answer briefly:

1. FAO (4)
2. Nutritional problems during adolescence (3)

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Asterisks (*) denotes repetition.

BIOCHEMISTRY

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Asterisks (*) denotes repetition.

UNIT-1

INTRODUCTION

Short essay:

1. Discuss the transport mechanisms across the cell membrane (5)

Short note:

1. Structure and functions of mitochondria (3)

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UNIT-2

METABOLISM OF CARBOHYDRATES

Essays:

1. What is the normal level of blood glucose. Describe the regulation of blood glucose (10)
2. Discuss the digestion and absorption of carbohydrates. Add a note on the metabolic disorders associated with carbohydrate digestion. (10)
3. Mention the normal value of blood glucose. Explain in detail about the regulation of blood glucose. Add a note on oral glucose tolerance test (10)
4. Mention the normal fasting blood sugar level. How is the blood sugar regulated*(10)
5. Write an essay of TCA cycle. Add a note on its energetic (7+3=10)

Asterisks (*) denotes repetition.

Short essay:

1. Digestion and absorption of carbohydrates* (5)
2. Classification and biological importance of carbohydrates (5)

Short notes:

1. Biochemical derangements in diabetes mellitus (5)
2. Benedict's test. (3)
3. What is the normal blood glucose level and how it is regulated (3)
4. Oral Glucose Tolerance Test* (5)

Difference between:

1. Starch and glycogen (3)
2. Hypoglycemia and hyperglycemia (3)

Answer briefly:

1. Disaccharides (3)

TCON ACADEMIC CLUB

Asterisks (*) denotes repetition.

UNIT-3

METABOLISM OF LIPIDS

Essay:

1. Classify lipoproteins. Describe the metabolism of very low density lipoprotein. Add a note on atherosclerosis. (10)
2. Name the ketone bodies. Explain how ketone bodies are synthesized and utilized in our body. Add a note on diabetic ketoacidosis. (10)

Short essays:

1. Mention the classifications of fatty acids and add a note on PUFA. (5)

Short notes:

1. Phenyl ketonuria (5)
2. Lipoproteins** (5)
3. Classification of lipids (5)
4. Atherosclerosis (5)
5. Ketogenesis (3)
6. Formation and utilization of ketone bodies* (5)
7. How lipids are digested and absorbed from the body* (5)
8. Phospholipids. (3)

Answer briefly:

1. Lipid profile. (3)
2. Diabetic ketoacidosis (3)

Difference between:

1. Saturated fatty acids and unsaturated fatty acids (3)
2. Diabetic ketoacidosis and starvation ketoacidosis (3)
3. Simple lipids and compound lipids (3)

Asterisks (*) denotes repetition.

TCON ACADEMIC CLUB

UNIT-4

METABOLISM OF AMINO ACIDS & PROTEINS

Essays:

1. What are the essential amino acids? Explain the digestion and absorption of proteins(10)
2. Discuss the formation and disposal of ammonia **(10)
3. Describe the biologically important compounds derived from phenyl-alanine. Mention the inborn errors associated with phenyl-alanine (10)
4. How is toxic ammonia converted to non-toxic urea. Explain the process.(5)

Short essay:

1. What is the normal blood urea level.. Discuss how ammonia is converted to urea (5)
2. Describe the classification of amino acids based on nutritional importance (5)

Short note:

1. Urea cycle (5)
2. Plasma Protein* (5)
3. Functions of plasma protein (3)
4. Biologically important substances synthesized from phenyl alanine (5)
5. Phospholipids. (5)

Differentiate between:

1. Essential amino acids and non-essential amino acids* (3)
2. Glucogenic amino acids and ketogenic amino acids (3)

Asterisks (*) denotes repetition.

UNIT-5

CLINICAL ENZYMOLOGY

Short essays:

1. Enumerate the enzymes used to assess hepatic injury (5)
2. What are isoenzymes. Give examples of isoenzymes with their clinical importance(3)

Answer briefly:

1. Enzymes elevated in acute myocardial infarction (3)

TCON ACADEMIC CLUB

UNIT-6

VITAMINS AND MINERALS

Essays:

1. Dietary sources, biochemical functions and deficiency manifestations of vitamin A**(10)
2. Explain the sources, factors affecting absorption and the deficiency manifestations of Iron. (10)
3. What is normal serum calcium level. How is it regulated. Add a note on hypercalcemia and hypocalcemia (10)
4. Explain the sources, biochemical functions and deficiency manifestations of vitamin C **(10)
5. Dietary sources, biochemical functions and deficiency manifestations of vitamin D *(10)

Asterisks (*) denotes repetition.

Short essay:

1. Discuss the biochemical role of thiamine(5)
2. Mention the daily requirement, dietary source and deficiency manifestations of vitamin D.(5)

Short note:

1. Functions of calcium (3)
2. Biochemical role of vitamin A (5)
3. Scurvy*(5)
4. RDA, dietary sources, biochemical features, and deficiency manifestations of vitamin C (5)
5. Regulation of blood calcium (5)
6. Sources and deficiency manifestations of vitamin B6 (5)
7. Write down the normal level of calcium. How is calcium regulated in the body (5)

Answer Briefly:

1. Pellagra (3)

Differentiate

1. Hypokalemia and hyperkalemia (3)

TCON ACADEMIC CLUB

Asterisks (*) denotes repetition.

UNIT-7

ACID BASE BALANCE

Short notes:

1. Metabolic acidosis (5)
2. Plasma buffers* (3)
3. Blood buffers (3)

Answer briefly:

1. Metabolic acidosis (3)

Differentiate between:

1. Metabolic acidosis and metabolic alkalosis (3)
2. Respiratory acidosis and respiratory alkalosis(3)

TCON ACADEMIC CLUB

UNIT-8

FUNCTION TEST

Asterisks (*) denotes repetition.

UNIT-9

HAEMOGLOBIN

Short note:

1. How heme is catabolized (5)
2. Catabolism of bilirubin (5)
3. Breakdown of hemoglobin (5)
4. Jaundice (5)

Differentiate between:

1. Prehepatic jaundice and obstructive jaundice (3)

TCON ACADEMIC CLUB

UNIT-10

IMMUNOCHEMISRY

Short note:

1. Structure of immunoglobulin (5)

Differentiate between:

1. Immunoglobulin(G) and immunoglobulin(M) *(3)

Asterisks (*) denotes repetition.

PSYCHOLOGY

TCON ACADEMIC CLUB

Asterisks (*) denotes repetition.

UNIT-1

INTRODUCTION

Essays:

1. Define psychology. Explain any three methods of psychology with its merits and demerits. (2+10)

Short notes:

1. Methods of psychology. (5)
2. Psychology related to nursing. (5)
3. Origin of the science of psychology. (5)
4. Introspection method. (5)

Answer briefly:

1. Observation method. (3)
2. Introspection.(3)
3. Case study method. (3)
4. Interview method. (3)
5. Historical background of psychology. (3)

Asterisks (*) denotes repetition.

UNIT-2

BIOLOGY OF BEHAVIOR

Essays

1. Explain body mind relationship with suitable examples. (12)
2. Define heredity. Explain the relation between brain and behavior. (2+10)

Short notes:

3. Autonomic nervous system and behavior. (5)
4. Brain and behavior. (5)
5. Body and mind relationship. (5)
6. Nervous system and behavior. (5)
7. Right and left hemisphere related to human behavior. (5)

Answer briefly:

1. Heredity and environment.* (3)
2. Heredity.(3)

Asterisks (*) denotes repetition.

UNIT-3

COGNITIVE PROCESS

Essays:

1. What is intelligence quotient? Explain the theories of intelligence(2+8)
2. Define thinking and explain various types of thinking*.(2+8)
3. Define memory. Discuss the types of memory, factors influencing memory and theories of forgetting.(2+8)
4. Define learning and explain classical and operant conditioning.(2+8)
5. Define learning. Discuss the factors influencing learning and theories of learning(2+8)
6. Define attention. Explain determinants and types of attention.(2+8)
7. What is learning? Explain in detail Pavlov's classical conditioning with its significance(2+8)

Short essays:

1. Factors influencing learning.(5)
2. Explain any FOUR intelligence tests in detail(5)
3. What is projective test? Describe the uses and limitations of projective technique*(5)
4. Explain basics of operant conditioning.(5)

Short notes:

1. Perceptual organization .(5)
2. Classical conditioning .(5)
3. Aptitude tests.(5)
4. Type of thinking.*(5)
5. Uses of intelligence tests(5)

Asterisks (*) denotes repetition.

6. Instrumental conditioning.(5)
7. Theory of insight full learning.(5)
8. Principles of perception.(5)
9. Reasoning and types of reasoning.(5)
10. Types of intelligence(5)
11. Factor affecting perception(5)
12. Mention the types of memory. Explain the long term memory.(5)
13. Principles of perceptual organization (5)
14. Assessment of intelligence(5)
15. Factors influencing attention.(5)
16. Errors of perception(5)
17. Methods to improve memory.(5)
18. Explain Spearman's two factor theory of intelligence.(5)
19. Adaptation and coping with stress(5)
20. Explain Pavlov's conditioning theory in learning.(5)
21. Describe the methods of memorization.(5)
22. Laws of learning.(5)
23. Levels of thinking.*(5)
24. Psychoanalytical theory of personality.(5)
25. Effective methods of memorization.(5)
26. Stages in creative thinking(5)
27. Theories of forgetting.(5)
28. Errors in Perception.(5)
29. Subjective factors determining attention.(5)
30. Nature of learning(5)

Answer briefly:

1. Types of Attention*.(3)
2. Aptitude.(3)
3. Types of aptitudes (3)
4. Illusion(3)
5. Study habit.(3)
6. Hallucination..(3)

Asterisks (*) denotes repetition.

7. Intelligence quotient.* (3)
8. Nature of attitude.(3)
9. Span of attention.(3)
10. Mental retardation.(3)
11. Transfer of learning(3)
12. Factors affecting learning (3)
13. Determinants of attention (3)
14. Errors in perception (3)

TCON ACADEMIC CLUB

Asterisks (*) denotes repetition.

UNIT-4

MOTIVATION, EMOTION AND STRESS

Essays:

1. Define emotion. Explain the physiology of emotion(2+8)
2. Explain emotion and explain its components in detail(2+8)
3. Define emotions. Discuss the theories of emotions and emotion in health & illness.(2+5+3)
4. Give the concept of need, motive and drive. Explain theories of motivation. (12)
5. What is motivation? Explain various types of motives.(2+8)
6. Describe the psychometric assessments of emotions and attitudes(10)
7. Explain stress in detail.(10)

Short essays:

1. Define motivation. Explain the primary motives.(7)
2. Explain the theories of emotions(7)

Short notes:

1. Explain physiological changes that take place during emotion.(5)
2. Elaborate theories of emotion.(5)
3. Techniques to reduce stress.(5)
4. Types of social motive.(5)
5. Theories of emotions.*(5)
6. Explain primary motives(5)
7. Adaptation and coping with stress.(5)

Asterisks (*) denotes repetition.

8. Enumerate behavioral changes during stress.(5)
9. Cannon Bard theory of emotion.(5)
10. Explain the physiological changes that occur during an emotion(5)
11. Drive theory of motivation (5)

Answer briefly:

1. Physiological Motives.(3)
2. Types of conflict.(3)
3. Stress.(3)
4. Motivation(3)
5. Reasons for stress.(3)
6. Maslow theory of motivation.(3)

TCON ACADEMIC CLUB

Asterisks (*) denotes repetition.

UNIT-5

PERSONALITY AND ATTITUDE

Essays:

1. Define personality. Discuss the theories of personality and assessment of personality(10)

Short notes:

1. Objective assessment of personality.(5)
2. Explain factors influence attitudes.(5)
3. Types of personality.* (5)
4. Explain the topography of personality.(5)
5. Mental hygiene.(5)
6. Elaborate Freud's psycho analytical theory.(5)
7. Psychosexual development.(5)

Answer briefly:

1. Development of attitude.(3)
2. Theories of personality.(3)
3. Attitude change.(3)
4. Extravert personality.(3)
5. Rorschach ink blot test.(3)

Asterisks (*) denotes repetition.

UNIT-6

DEVELOPMENTAL PSYCHOLOGY

Short essay:

1. Describe classical conditioning with example (7)

Short notes:

1. Types of social groups.(5)
2. Psychology of adulthood.(5)
3. Qualities of a group leader.(5)

Answer briefly:

1. Infancy(3)
2. Prenatal period (3)

TCON ACADEMIC CLUB

Asterisks (*) denotes repetition.

UNIT-7

MENTAL HYGIENE AND MENTAL HEALTH

Essays:

1. What is defense mechanism and explain various defense mechanism in detail.(2+8)
2. Define mental hygiene. Describe the concepts of mental hygiene and mental health.(12)

Short essays:

1. Describe the role of nurses in mental health.(7)

Short notes:

1. Characteristics of mental health*(5)
2. Causes of frustration.(5)
3. Describes the characteristics of mental health.(5)
4. Prevention of mental illness(5)
5. Principles in counseling.(5)
6. Describe steps in conflict resolution.(5)
7. Explain steps in counseling(5)
8. Explain nurse's role in guidance and counseling(5)
9. Defense mechanisms (5)
10. Role of nurse in mental health (5)

Answer briefly:

1. Old-age health(3)
2. In-group and out-group.(3)
3. Health and mental health.(3)

Asterisks (*) denotes repetition.

4. Counselling(3)
5. Characteristics of mentally healthy person.(3)
6. Warning signs of poor mental health.(3)
7. Characteristics of mentally healthy person.(3)
8. Describe the concept of mental health(3)
9. Types of guidance.(3)
10. Warning signs of mental illness.(3)

TCON ACADEMIC CLUB

Asterisks (*) denotes repetition.

UNIT 8

PSYCHOLOGICAL ASSESSMENT AND TEST

Essays:

1. Define personality. Explain the tests of personality(2+8)

Short essays:

1. Psychological tests. (7)
2. Explain role of nurse in psychological assessment (7)

Short notes:

1. Projective techniques. (5)
2. Bhatia's performance test.(5)

Answer briefly:

1. Enumerate any six standard psychological tests.(3)
2. Thematic Apperception Test (TAT). (3)
3. Uses of psychological test. (3)

Asterisks (*) denotes repetition.

NURSING FOUNDATION

TCON ACADEMIC CLUB

Asterisks (*) denotes repetition.

UNIT-1

INTRODUCTION

Essays:

1. Define health. Explain the levels of disease prevention. Describe the impact of illness on the patient and family. (10)
2. Define illness. Enlist the risk factors for developing illness. Describe the stages of illness behavior (10)
3. Define health and illness. Discuss the various levels of prevention. List the type of health care agencies (10)

Short notes:

1. Dimensions of health (5)
2. Types of health care agencies (5)
3. Health illness continuum (5)
4. Levels of prevention (5)
5. Nurses role in health promotion (5)
6. Stages of illness behavior (5)
7. Code of ethics (5)
8. Barriers of communication (5)
9. Levels of health care (5)
10. Factors influencing health. (5)
11. Universal safety precaution (5)
12. Health care team (5)
13. Illness behavior (5)

Asterisks (*) denotes repetition.

Differentiate between:

1. Primary prevention and secondary prevention ** (4)
2. Health and illness * (4)
3. Sign and symptom (3)

List down:

1. Four characteristics of a profession (4)
2. Six functions of hospital *** (3)
3. Types of bed (3)

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Asterisks (*) denotes repetition.

UNIT-2

NURSING AS A PROFESSION

Essays:

1. Define profession. Explain nursing as a profession. Describe the values in professional nursing (10)
2. Define nursing. Explain the roles of a professional nurse. Describe Orem's self care deficit theory. (10)
3. Define the term profession. List the characteristics of profession and justify nursing as a profession (10)
4. Define oxygen inhalation and mention the indications of oxygen inhalation. Explain the role of nurse in oxygen administration (10)

Short notes:

1. Categories of nursing personnel (5)
2. Criteria of a profession (5)
3. Qualities of a nurse * (5)
4. Characteristics of a profession* (5)
5. Code of ethics. (5)
6. Functions of a nurse (5)

Differentiate between:

1. Autonomy and accountability (3)
2. Negligence and malpractice (3)
3. Non maleficence and beneficence * (4)
4. Subjective and objective data ** (4)
5. Empathy and sympathy ** (3)
6. Ethics and values (3)

Asterisks (*) denotes repetition.

List down:

1. Four characteristics of a profession (4)
2. Six ethical principles in nursing (3)
3. Six qualities of a nurse * (3)
4. Six characteristics of nursing profession (3)
5. Six processes which helps to clarify the values (3)

UNIT-3

CONCEPTUAL AND THEORETICAL MODELS IN NURSING PRACTICE

Short essays:

1. Explain Maslow's hierarchy of needs * (10)

Short notes:

1. Health belief model (5)
2. Betty Newman's system theory (5)
3. Maslow's hierarchy of needs ** (5)

List down:

1. Six nursing theorists and their theories (3)

Asterisks (*) denotes repetition.

UNIT-4

COMMUNICATION & NURSE PATIENT RELATIONSHIP

Essays:

1. Define 'communication'. Describe the elements of communication. Explain the factors influencing communication (1+4+5=10)
2. Define 'communication'. Describe the elements of communication. Explain the factors influencing communication (1+4+5=10)
3. Define communication. Discuss the phases of helping relationship and barriers to effective communication (2+4+4=10)
4. Define communication. Explain the elements of communication. Describe the different types of communication (1+4+5=10)

Short essays:

1. Define communication and mention the types of communication. Explain the methods of effective communication (10)

Short notes:

1. Levels of communication ** (5)
2. Phases of helping relationship ** (5)
3. Barriers of communication **** (5)
4. Purposes of patient teaching (5)
5. Elements of communication (5)
6. Interpersonal relationship (5)
7. Communication (5)

Asterisks (*) denotes repetition.

Differentiate between:

1. Verbal and non-verbal communication (3)

List down:

1. Four therapeutic communication techniques (4)
2. Four phases of helping relationship (4)
3. Six barriers to communication (3)
4. Six basic elements of communication process * (3)
5. Levels of communication (3)

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Asterisks (*) denotes repetition.

UNIT-5

HEALTH ASSESSMENT

Essays:

1. Define nursing process. Explain the characteristics of nursing process. Describe the nursing management of an unconscious patient based on nursing process approach. (1+4+5=10)
2. Define physical examination. Explain the methods of physical examination. Describe the nurse's role in physical examination (1+4+5=10)
3. Define nursing diagnosis and describe the types of nursing diagnosis. Explain the components of NANDA nursing diagnosis with suitable examples (1+4+5=10)

Short essays:

1. Explain planning as a step of nursing process (10)
2. Define nursing process. Explain the methods of health assessment. Discuss the planning phase of nursing process (1+4+5=10)
3. Nursing diagnosis (5)

Short notes:

1. Nurse's role in physical examination (5)
2. Assessment phase of nursing process (5)
3. Components of nursing health history (5)
4. Steps of nursing process (5)
5. Components of a nursing care plan (5)
6. Nursing diagnosis (5)
7. Evaluation in nursing process (5)
8. Planning phase of nursing process (5)
9. Methods of physical examination (5)

Asterisks (*) denotes repetition.

10. Nursing assessment (5)
11. Types of nursing interventions (5)
12. Techniques of physical examination (5)

Differentiate between:

1. Goals and expected outcomes (3)
2. Medical and nursing diagnosis (4)
3. Inspection and auscultation (4)
4. Nursing diagnosis and medical diagnosis (3)
5. Palpation and Percussion ** (3)
6. Medical diagnosis and nursing diagnosis * (3)

List down:

1. Six positions for physical examination (3)
2. Methods of physical examination (3)
3. Four methods of health assessment (4)
4. Six guidelines for writing nursing diagnostic statement (3)
5. Types of care plan (3)

TCON ACADEMIC CLUB

Asterisks (*) denotes repetition.

UNIT-6

DOCUMENTATION & REPORTING

Essays:

1. Define documentation. List the purposes of recording and reporting. Explain the general guidelines for documentation. Describe the types of documentation system. (1+2+5+7=15)

Short notes:

1. Sources of data (5)
2. Guidelines for documentation * (5)
3. Types of records (5)
4. Importance of records (5)
5. Characteristics of a good report (5)

Differentiate between:

1. Autonomy and accountability (3)
2. Change of shift report and incident report (3)
3. Incident report and transfer report (4)
4. Records and reports * (4)

List down:

1. Six purposes of documentation.* (3)
2. Six guidelines for documentation * (3)
3. Six methods of documentation in nursing (3)
4. Four values of records (4)

Asterisks (*) denotes repetition.

UNIT-7

HOSPITAL ADMISSION AND DISCHARGE

Short essays:

1. Discharge procedure * (5)
2. Types of hospital admission (5)
3. Medico legal issues in hospital admission (5)
4. Admission procedure * (5)
5. Principles of admission procedure (5)
6. Types of discharge from the hospital (5)

List down:

1. Six purposes of admission procedure (3)
2. Types of admission (3)
3. Six functions of hospital * (3)
4. Six types of discharge from hospital (3)

TCON ACADEMIC CLUB

Asterisks (*) denotes repetition.

UNIT-8

INFECTION CONTROL AND WASTE MANAGEMENT

Essays:

1. Define health care associated infections. Describe the types of health care associated infections. Explain the causes of health care associated infections.(1+4+5=10)
2. Define infection and describe the modes of transmission of infections. Explain the body defences against infections (1+4+5=10)

Short essays:

1. Define communication and mention the types of communication. Explain the methods of effective communication (1+3+6=10)
2. List the types of hospital wastes. Describe the hazards associated with hospital wastes (4+6=10)

Short notes:

1. Nosocomial infection * (5)
2. What are the universal safety precautions (5)
3. Universal safety precautions * * (5)
4. Transmission based precautions (5)
5. Prevention of cross infection in the hospital (5)
6. Standard infection precautions* (5)
7. Biomedical waste management (5)
8. Chain of infection (5)

Asterisks (*) denotes repetition.

Differentiate between:

1. Standard precautions and isolation precautions (3)
2. Medical asepsis and surgical asepsis *** (3)
3. Passive and active immunity (4)
4. Concurrent and terminal disinfection * (4)
5. Infection and inflammation (4)
6. Disinfection and sterilization * (3)
7. Source isolation and protective isolation* (3)

List down:

1. Six factors influencing efficacy of sterilization (3)
2. Six principles of surgical asepsis (3)
3. Types of hospital waste * (3)
4. Four principles of Biomedical waste management (4)
5. Four types of personal protective equipment (4)
6. Six standard precautions for infection control (3)
7. Six methods of hospital waste management (3)
8. Six principles of surgical asepsis (3)

TCON ACADEMIC CLUB

Asterisks (*) denotes repetition.

UNIT-9

VITAL SIGNS

Essays:

1. Define respiration. Explain the characteristics of respiration. Describe the factors affecting respiration (1+4+5=10)
2. Mention the factors affecting oxygenation. List the complications of oxygen administration. Discuss the nursing care of patients receiving oxygen administration (2+3+5=10)
3. Define fever. Describe the factors affecting body temperature. Explain the nursing care of a patient with fever (15)
4. Define pulse and explain the factors affecting pulse. Describe the characteristics of pulse (1+5+4=10)
5. Define pulse. Explain the characteristics of pulse. Describe the factors affecting pulse (1+4+5=10)
6. Define blood pressure and describe the physiology of arterial blood pressure. Explain the factors affecting blood pressure. (1+4+5=10)
7. Define pressure ulcer. Enumerate the common sites of pressure ulcer. State the risk factors of pressure ulcer. Explain the nursing interventions for the prevention of pressure ulcer. (15)

Short essays:

1. Explain the factors affecting the pulse (10)
2. Define catheterization. What are the indications of catheterization. Explain the nursing management of a client with continuous bladder drainage. (7)

Asterisks (*) denotes repetition.

Short notes:

1. Guidelines for measuring vital signs (5)
2. Sites for temperature measurement (5)
3. Management of hyperthermia (5)
4. Characteristics of pulse (5)
5. Factors affecting blood pressure.** (5)
6. Factors affecting body temperature * (5)
7. Sites for assessment of pulse (5)
8. Factors influencing character of respirations (5)
9. Factors affecting heat production in the body (5)
10. Care of patients with dyspnea (5)
11. Factors affecting pulse rate* (5)
12. Alterations in respiratory functioning (5)

Difference between:

1. Tachycardia and tachypnoea (3)
2. Orthopnoea and apnoea (3)
3. Diastolic and systolic blood pressure (3)
4. Tachycardia and bradycardia ***** (3)
5. Blood pressure and pulse pressure (4)
6. Systolic and diastolic pressure * (4)
7. Hyperthermia and hypothermia * (4)
8. Dyspnoea and orthopnoea (3)
9. Eupnea - Apnea. (3)
10. Dicrotic pulse and bigeminal pulse (3)
11. Hypotension and hypertension (3)
12. Pulse deficit and pulse pressure (3)
13. Osmotic pressure and hydrostatic pressure (3)
14. Isometric exercise and isotonic exercise (4)
15. Gastric gavage and gastric lavage (4)

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Asterisks (*) denotes repetition.

List down:

1. Six sites for checking body temperature (3)
2. Six sites for assessing pulse (3)
3. Four factors affecting pulse (4)
4. Six sites for checking peripheral pulses * (4)
5. Four alterations in respiration (4)
6. Four abnormal pulses (4)
7. Characteristics of pulse (3)
8. Six factors influencing blood pressure (3)
9. Six factors affecting blood pressure (3)
10. Four positions and its purposes. (4)
11. Four comfort devices and its purposes. (4)

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Asterisks (*) denotes repetition.

UNIT-10

MEETING NEEDS OF PATIENT

Essays:

1. Mention the factors affecting oxygenation. List the complications of oxygen administration. Discuss the nursing care of patients receiving oxygen administration (2+3+5=10)
2. List the types(methods) of oxygen administration. Explain the nurses responsibility in administration of oxygen * (5+10=15)
3. Define urinary retention. Explain the principles of urinary catheterization. Describe the nurse's role in urinary catheterization * (1+4+5=10)
4. Define pressure ulcer. Write the common sites of pressure ulcer. Describe Nursing care of pressure ulcer. (2+2+6=10)
5. Explain the factors affecting medication administration. Describe the procedure for administration of subcutaneous injection.(4+6=10)
6. Describe the factors affecting hygiene. Explain the care of pressure point (5+5)
7. Define Ryles tube feeding and enumerate the principles of Ryles tube feeding. Explain the role of nurse in Ryles tube feeding (1+4+5=10)
8. Define first aid and enumerate the qualities of a first aider. Explain the first aid to a client with food poisoning (1+4+5=10)

Short essays:

1. Explain the nurses' responsibility in performing bowel wash (10)
2. Define enema. Explain the procedure for giving enema to a patient (10)
3. Define acid base balance. What are the factors affecting fluid electrolyte balance. Explain the nursing interventions in fluid electrolyte imbalance (1+4+5=10)

Asterisks (*) denotes repetition.

Short notes:

1. Systemic effects of immobility (5)
2. Factors affecting bowel elimination (5)
3. Medication errors (5)
4. Methods of oxygen delivery (5)
5. Prevention of infection in patients with urinary catheters (5)
6. Sleep disorders ** (5)
7. Principles of administering enema. (5)
8. Nursing management of insomnia (5)
9. Spiritual care (5)
10. Enema (5)
11. Spiritual needs of the patient * (5)
12. Nebulisation (5)
13. Body mechanics (5)
14. Constipation and its management (5)
15. Care of unconscious patient (5)
16. Management of constipation * (5)
17. Nurses responsibility in blood transfusion (5)
18. Characteristics of urine (5)
19. Nursing interventions for a patient with fluid volume excess (5)
20. Positioning bedridden clients (5)
21. Prevention of pressure ulcer (5)
22. Factors affecting therapeutic environment. (5)
23. Principles involved in tube feeding (5)
24. Nursing management of a client with indwelling catheter (5)
25. Types of position and its purpose * (5)
26. Care of patient with immobiiity (5)
27. Fluid and electrolyte imbalance (5)
28. Test for albumin in urine (5)
29. Purposes of urinary catheterization (5)
30. Care of urinary drainage (5)
31. Nurse's role in collection of specimen (5)

Asterisks (*) denotes repetition.

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32. Steam inhalation (5)
33. Nursing measures to improve appetite (5)
34. Nutritional assessment (5)
35. Risk factors for development of pressure ulcers (5)
36. Chain of infection (5)

Differentiate between:

1. Acidosis and alkalosis (3)
2. Respiratory acidosis and respiratory alkalosis (3)
3. Supine position and sims' position (3)
4. Algor mortis and Livor mortis. (3)
5. Oliguria and anuria (3)
6. Colostomy and gastrostomy (3)
7. Active exercise and passive exercise * (4)
8. Urinary retention and urinary incontinence (4)
9. Instillations and irrigations (4)
10. Empathy and sympathy (3)
11. Constipation and faecal impaction (3)
12. Insomnia – Parasomnia (3)
13. Rigor mortis - Algor mortis * (3)
14. Stomatitis and glossitis (3)
15. Isotonic exercise and isometric exercise (3)
16. Constipation and diarrhea (3)
17. Autopsy and biopsy * (3)
18. Lithotomy position and trendelenburg position (3)
19. Dysphagia and dysuria (3)
20. Constipation and faecal impaction (3)
21. Sign and symptom (3)
22. Isometric contraction and isotonic contraction (3)
23. Osmotic pressure and hydrostatic pressure (3)

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Asterisks (*) denotes repetition.

List down:

1. Six factors influencing hygiene* (3)
2. Six effects of exercise (3)
3. Six factors affecting sleep (3)
4. Four types of beds * (3)
5. Four types of oxygen delivery devices (4)
6. Four indications for oxygen administration (4)
7. Four factors influencing urination (4)
8. Four types of enema (4)
9. Six common problems of bowel elimination (3)
10. Six indications of gastric lavage (3)
11. Six measurers to improve appetite (3)
12. Principles of back care * (3)
13. Factors affecting nutritional needs (3)
14. Six factors affecting oxygenation (3)
15. Types of comfort devices * (3)
16. Complications of prolonged immobility * (3)
17. Types of bed * (3)
18. Six positions and its purposes (3)
19. Six complications of neglected mouth (3)

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Asterisks (*) denotes repetition.

UNIT-11

THERAPEUTIC USE OF HEAT AND COLD APPLICATION

Short notes:

1. Therapeutic effect of cold (5)
2. Nurse's role in hot water bag application (5)
3. Sitz bath (5)

Differentiate between:

1. Cold compress and tepid sponge (3)
2. Hot and cold application (3)

List down:

1. Four principles of medicine administration. (4)

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Asterisks (*) denotes repetition.

UNIT-12

ADMINISTRATION OF MEDICATION

Essays:

1. Define drug administration and mention the various routes of drug administration. What are the factors influencing drug action. Explain the nurse's responsibility in storage and maintenance of drugs in the hospital (1+3+4+7=15)
2. Mention the rights of medication administration. Explain the procedure of intramuscular medicine administration. * (5+10=15)
3. Define medication. Explain the components of medication order. Describe the sites of intramuscular injections (1+4+5=10)

Short essay:

1. Explain the nurses responsibility in the administration of oral medication (10)

Short notes:

1. Medication errors (5)
2. Nurses' responsibility in storage and maintenance of drugs (5)
3. Complications of intravenous therapy (5)
4. Sites for intramuscular injection (5)
5. Intra dermal injection (5)
6. Factors affecting medication action (5)
7. Responsibilities of nurse in intravenous infusion (5)
8. Complications of injections (5)

Asterisks (*) denotes repetition.

Differentiate between:

1. Protocols and standing orders (3)
2. Drug tolerance and drug interaction (3)
3. Intradermal injection and subcutaneous injection (4)
4. Side effects and adverse effects (3)
5. PRN order and SOS order (3)
6. Innunction and instillation (3)
7. Instillation – Inhalation (3)
8. Injection and infusion (4)

List down:

1. Six factors influencing drug actions (3)
2. Six complication of enteral feeding (3)
3. Six routes of medication administrations (3)
4. Six indications for enteral nutrition (3)
5. Routes of drug administration * (3)
6. Six essential parts of a medication order (3)
7. Six advantages of oral medication *(3)

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Asterisks (*) denotes repetition.

UNIT-13

REHABILITATION

Short note:

1. Types of rehabilitation (5)
2. Role of nurse in rehabilitation** (5)
3. Rehabilitation process (5)

Differentiate between:

1. Rehabilitation and continuing care (3)

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UNIT-14

CARE OF TERMINALLY ILL PATIENT

Short notes:

1. Care of a dying Patient (5)
2. Signs of clinical death (5)
3. Grieving Process (5)
4. Care of dead body. (5)

Differentiate between:

1. Grief and bereavement (3)

Asterisks (*) denotes repetition.

UNIT-15

FIRST AID NURSING

Short essays:

1. Define first aid and enumerate the qualities of a first aider. Explain the first aid to a client with food poisoning* (1+4+5=10)

Short notes:

1. First aid in electrocution (5)
2. Cardiopulmonary resuscitation (5)
3. First aid in burns (5)
4. Principles of bandaging * (5)
5. First aid measures of a patient with snake bite (5)
6. First aid in drowning (5)
7. First aid of fracture of lower limb (5)

List down:

1. Six qualities of a first aider * (3)
2. Four first aid measures in dog bite (4)

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Asterisks (*) denotes repetition.